

Tuning to Terahertz Electronics



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The terahertz (THz) region (nominally 0.1-10THz) separates electronics from photonics, and has historically been difficult to access. Semiconductor electronics run out of steam after $\sim 100\text{GHz}$ due to transport time limitations. Photonic devices falter below $\sim 10\text{THz}$ because photon energy drops to thermal energy. At the high-frequency end, infra-red (IR) optoelectronics cannot operate significantly below 10THz . Since these devices employ photon-electron particle interactions, as photon energy $h\nu$ decreases below thermal energy kT , the device ceases to operate efficiently unless it is cooled down. This adds significant cost and weight. At the low-frequency end, semiconductor electronic devices cannot operate at frequencies significantly above 100GHz . Transport time across the semiconductor junction is limited by drift and diffusion speeds. The largely-unapped frequency region between 100GHz and 10THz (the THz region) holds promise for a wide range of commercial and military applications. Terahertz electronics (TE) is a new technology that extends the range of electronics into the THz-frequency region.

Terahertz electronics

Terahertz electronics technology opens up practical applications in high-speed data interconnects, THz imaging, and highly-integrated radar and communication systems. The gap between electronics and photonics has closed. Further use and development of such technological

devices will make TE a reality in the near future. It does not use semiconductors; instead, it is based on metal-insulator tunnelling structures to form diodes for detectors and ultra-high-speed tunnelling transistors for oscillator based transmitters. With these devices, detectors and transistors for operation in the THz region have been designed. Besides being extremely fast, TE devices are made entirely of thin-film materials—metals and insulators—and so may be fabricated on top of complementary metal oxide semiconductor (CMOS) circuitry—a technology for constructing integrated-circuits circuitry or on a wide variety of substrate materials.

Based on metal-insulator tunnel junctions, TE technology extends the range of electronics beyond the 100GHz barrier to 10THz . In these devices, charge transport through the junction occurs via electron tunnelling, which has a time constant of $\sim 10^{-15}$ seconds. Charge transport to and from the junction occurs via plasma oscillation in metals, which is easily supported with low loss in the THz range of interest. Furthermore, epitaxial growth and high process temperatures are not required. Thus, integration of TE onto low-loss insulating substrates (for example, glass, sapphire, ceramic or plastic) or onto silicon-integrated circuits is possible. Industries are employing this concept in practical components, like detectors, transistors, mixers and other devices, to provide a full suite of ultra-fast integrated components for building high-frequency electromagnetic wave circuits and systems. The result is complete integrability, and consequently, low cost. In addition, devices may be fabricated onto large-area flat panels or flexible sheets, enabling completely integrated microwave/millimetre-wave/sub-millimetre-wave sensor and

Fig. 1: Terahertz gap

